## 3.1.7 Digital design and manufacture

### Computer aided design (CAD)

| Content | Potential links to maths and science |
| --- | --- |
| Students should be aware of, and be able to describe, the following:<br>• the advantages and disadvantages of using CAD compared to a manually generated alternative<br>• the use of CAD to develop and present ideas for products, including:<br>  • the use of 2D CAD for working drawings<br>  • the use of 3D CAD to produce presentation drawings<br>• how CAD is used in industrial applications. | Use of datum points and geometry when setting out design drawings.<br>The use of tolerances in dimensioning. |

### Computer aided manufacture (CAM)

| Content | Potential links to maths and science |
| --- | --- |
| Students should be aware of, and be able to describe, how CAM is used in the manufacture of products.<br>Specific processes to include:<br>• laser cutting<br>• routing<br>• milling<br>• turning<br>• plotter cutting. | Calculating speeds and times for machining. |

## Virtual modelling

| Content | Potential links to maths and science |
|---|---|
| Students should be aware of, and be able to describe, how virtual modelling/testing is used in industry prior to product production.<br>Specific processes to include:<br>• simulation<br>• computational fluid dynamics (CFD) as used for testing aerodynamics and wind resistance, and flow of liquids within/ around products<br>• finite element analysis (FEA) as used in component stress analysis. | Interpretation of data from CFD or FEA testing. |

## Rapid prototyping processes

| Content | Potential links to maths and science |
|---|---|
| Students should be aware of, and be able to describe, rapid prototyping processes, including 3D printing.<br>Students should understand, and be able to explain, the benefits to designers and manufacturers. | Calculating volumes of 3D printed products, calculating time/speed for 3D printing. |

## Electronic data interchange

| Content | Potential links to maths and science |
|---|---|
| Students should be aware of, and able to describe, the use of electronic point of sales (EPOS) for marketing purposes and the collection of market research data, including:<br>• the maintenance of stock levels<br>• the capture of customer data, eg contact details. | |

## Production, planning and control (PPC) networking

| Content | Potential links to maths and science |
|---|---|
| Students should be aware of, and able to describe, the role of PCC systems in the planning and control of all aspects of manufacturing, including:<br>• availability of materials<br>• scheduling of machines and people<br>• coordinating suppliers and customers. | |